

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 1 – Research Paper Review / Literature Survey

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO5	Assessment:	Assessment Tool 1 – Research Paper Review / Literature Survey
Weightage:	20%	Total Marks:	25
CO5 Statement:	Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly		
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Date:	24/08/2026	Duration:	60 minutes

Code of Conduct

I Tamil Elakiya S (192524443) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Tamil Elakiya S

Rubrics for Calculation

Criteria	Mark
Selection & Review of Research Papers(25%)	/5
Comparative Analysis(20%)	/5
Critical Analysis & Research Gap(25%)	/5
Future Directions & Relevance(15%)	/5
Documentation, Citations & References(15%)	/5
TOTAL	/25

Digital Farmer Marketplace: A Digital Platform for Direct Farmer–Buyer Connectivity

1. INTRODUCTION

Agriculture is one of the major sectors supporting the Indian economy and providing livelihood opportunities to a large population. However, farmers frequently face challenges in marketing their agricultural products. Traditional agricultural marketing systems commonly involve multiple intermediaries between farmers and final consumers. This can reduce the farmer's share of the final selling price and create difficulties in obtaining accurate information about market demand and prevailing prices. The growth of information and communication technologies has created opportunities to improve agricultural marketing through digital platforms. E-commerce, mobile applications, online marketplaces, digital payment systems, and real-time information services can connect farmers with buyers more directly. Such systems can improve market accessibility, product visibility, price transparency, and communication between participants.

Research conducted in India has already demonstrated the potential of digital technologies in agricultural marketing. Studies have examined farmer adoption of e-commerce, online agricultural auctions, e-NAM, digital communication tools, agricultural e-commerce platforms, and digital farmgate systems. However, existing research also identifies several challenges, including limited digital literacy, inadequate infrastructure, fragmented platforms, logistics problems, trust issues, and lack of interoperability.

The proposed Digital Farmer Marketplace addresses this problem by providing a centralized digital platform through which farmers can list agricultural products and buyers can discover available products directly. The system aims to simplify agricultural product trading and improve farmer–buyer connectivity through a user-friendly digital marketplace.

2. PROBLEM STATEMENT

Traditional agricultural marketing systems often create difficulties for farmers in reaching suitable buyers and obtaining transparent information about product prices and demand. Farmers may depend on intermediaries for selling their produce, which can increase the distance between producers and consumers.

In addition, small and marginal farmers may have limited access to digital technologies or may find existing digital agricultural platforms difficult to use. Agricultural products also have specific requirements related to freshness, availability, transportation, and timely delivery.

Therefore, there is a need for a simple, accessible, and agriculture-specific digital marketplace that directly connects farmers and buyers while providing product information, transparent pricing, search facilities, and an organized transaction process.

3. OBJECTIVES OF THE LITERATURE SURVEY

The objectives of this literature survey are:

1. To study existing digital technologies used for agricultural marketing.
2. To analyze existing farmer-to-buyer and agricultural e-commerce platforms.
3. To identify methodologies and technologies used in previous research.
4. To study the role of e-commerce and mobile technologies in improving farmer market access.
5. To compare existing approaches based on their methodology, technology, findings, and limitations.
6. To identify challenges related to digital literacy, infrastructure, logistics, trust, and platform interoperability.
7. To identify research gaps in existing digital agricultural marketplace solutions.
8. To propose possible improvements and future research directions for a Digital Farmer Marketplace.

4. LITERATURE SEARCH METHODOLOGY

The literature survey was conducted using reputed scholarly databases and academic search platforms.

Databases Used

- Google Scholar
- ScienceDirect
- SpringerLink
- Wiley Online Library
- Frontiers
- Other academic journal repositories

Search Keywords

The following keywords were used to identify relevant research:

- "Digital Farmer Marketplace"
- "Agricultural E-commerce"
- "Farmer Buyer Digital Platform"
- "Digital Agricultural Marketplace"
- "Farmer-to-Consumer Marketplace"
- "e-NAM Farmer Market Access"
- "Digital Platforms for Farmers"
- "Agricultural Product E-commerce"
- "Digital Agriculture Marketing"
- "Smallholder Farmers Digital Market"

Research papers were selected based on their relevance to agricultural marketing, digital marketplaces, e-commerce adoption, farmer market access, and digital technologies.

5. REVIEW OF EXISTING RESEARCH PAPERS

The study examines the adoption of e-commerce practices among farmers in Trichy district [1]. The findings indicate that e-commerce can provide farmers with wider market access, better price information, and opportunities to reduce dependence on traditional intermediaries. However, limited awareness, inadequate technological knowledge, poor internet access, and farmers' hesitation to use digital platforms can restrict adoption. The study highlights the importance of developing simple, accessible, and farmer-friendly e-commerce systems that can support agricultural marketing and improve direct connections between farmers and buyers.

The research investigates the use of online auction systems for agricultural procurement, using coffee auctions in India as a case study [2]. The research shows that online procurement platforms can connect producers with a wider group of buyers, improve price discovery, and increase transparency in agricultural transactions. However, the effectiveness of such systems depends on internet access, digital literacy, reliable product information, and the participation of all relevant supply-chain stakeholders. Since the study mainly focuses on auction-based procurement, it does not provide a complete farmer-to-buyer marketplace with product listings, communication, payment, and delivery management. This creates an opportunity for developing a more comprehensive Digital Farmer Marketplace.

The research examines how cloud computing can support the operation of agricultural e-commerce platforms [3]. The study explains that cloud-based systems can manage product information, supply and demand data, online purchasing, payment processing, and delivery-related activities. Such platforms can connect farmers, traders, retailers, and consumers through a centralized digital marketplace. However, agricultural products have special requirements, including seasonal availability, perishability, quality variation, and transportation difficulties. Therefore, an agricultural e-commerce platform must provide features specifically designed for agricultural products rather than simply applying the structure of a general online shopping system.

The research studies the role of digital technologies in local agricultural and food systems [4]. The research indicates that digital farmgate platforms can improve communication, market access, information sharing, and coordination between farmers and buyers. However, the study identifies fragmentation and poor interoperability as major challenges because different platforms may use separate systems, data formats, and operating methods. This can make it difficult for farmers and other stakeholders to exchange information efficiently. The findings emphasize the importance of developing interoperable digital platforms that can support future integration with logistics providers, payment systems, agricultural databases, and other marketplace services.

The study examines how farmers used digital communication tools to market their agricultural products during the COVID-19 lockdown period [5]. The research identifies the use of mobile phones, telephone calls, SMS, WhatsApp, and Telegram for contacting buyers, sharing product information, and arranging sales. The findings demonstrate that farmers can use digital tools to maintain market connections when traditional marketing channels are disrupted. However, general-purpose communication applications do not provide structured product catalogues, organized buyer searches, price comparison, transaction records, or delivery management. The study therefore supports the development of a dedicated agricultural marketplace that combines communication, product listing, buyer discovery, and transaction-related services in one platform.

The research investigates the factors that influence small farmers' access to agricultural markets, including digital platforms such as e-NAM [6]. The study considers factors such as digital literacy, awareness, smartphone knowledge, access to information, education, and farmers' understanding of online applications. The findings indicate that farmers who are more familiar with smartphones and digital services are more likely to use online agricultural platforms. However, lack of awareness, limited technical

knowledge, perceived risks, poor infrastructure, and regional differences can reduce adoption. These findings suggest that a Digital Farmer Marketplace should use a simple interface, clear instructions, regional-language support, and easy navigation so that farmers with different levels of digital experience can use the system effectively.

The study examines how e-commerce platforms can support the digital marketing and distribution of agricultural products [7]. The research explains that digital platforms can improve product visibility, expand market reach, provide better access to buyers, and potentially improve farmers' income by reducing unnecessary intermediaries. At the same time, the research identifies several challenges, including inadequate rural infrastructure, limited digital literacy, logistics problems, product quality concerns, and difficulties in building trust between farmers and buyers. The findings indicate that the success of an agricultural e-commerce platform depends not only on its technical design but also on its usability, reliability, delivery support, and ability to address the practical needs of rural users.

The research proposes a mobile application that connects farmers and buyers through real-time market information and matching services [8]. The application allows farmers to register, upload details of available agricultural products, specify prices, and provide information about quantity and location. Buyers can search for commodities, compare available products, and identify suitable farmers. The study demonstrates the usefulness of mobile applications for improving direct farmer-buyer communication and reducing information gaps. However, the system requires further development in areas such as secure payments, product verification, logistics coordination, user authentication, scalability, and real-world testing. These requirements are relevant to the proposed Digital Farmer Marketplace, which can extend the basic matching concept with additional marketplace and transaction features.

The study examines how farmers can use mobile applications to access market information, crop prices, weather updates, and agricultural advisory services through smartphones [9]. The findings indicate that mobile applications can help farmers make better marketing decisions, reduce dependence on intermediaries, and improve access to timely agricultural information. However, limited internet connectivity, low digital literacy, language barriers, and the lack of localized information can restrict effective usage. The study suggests that agricultural applications should provide simple navigation, regional-language support, reliable information, and features designed according to farmers' practical needs. These findings support the development of a user-friendly Digital Farmer Marketplace that provides localized product information, buyer connectivity, and easy access to market-related services.

The research investigates the use of blockchain technology to improve transparency, traceability, and trust in agricultural transactions [10]. The study explains that blockchain can record product details, ownership changes, prices, and transaction information in a secure and tamper-resistant manner. This can help reduce information asymmetry, prevent fraudulent activities, verify the movement of agricultural products, and improve consumer confidence. However, blockchain implementation requires suitable technical infrastructure, accurate data entry, user awareness, and cooperation among farmers, buyers, transporters, retailers, and other stakeholders. The study indicates that future agricultural marketplaces can gradually incorporate blockchain-based traceability and secure transaction features to improve trust, accountability, and efficiency throughout the farmer-to-buyer supply chain.

6. COMPARATIVE LITERATURE SURVEY

Author/ Year	Research Problem	Methodology	Key Findings	Limitations	Research Gap
2007	Agricultural procurement	Case Study	Online auctions improve market connectivity	Supply-chain dependency	Complete farmer-centric marketplace
2013	Farmer e-commerce adoption	Survey	E-commerce has agricultural potential	Older technology context	Modern mobile marketplace
2022	Agricultural e-commerce	Platform research	Supports product search and transactions	Logistics/freshness challenges	Agriculture-specific services
2023	Digital farmgate fragmentation	Literature + stakeholder research	Digital platforms provide business benefits	Poor interoperability	Standardized/interoperable platforms
2023	Farmer digital marketing	Field study	Mobile communication widely used	No dedicated marketplace	Integrated agricultural platform
2025	Small farmer market access	Survey + statistical analysis	Digital literacy affects adoption	Awareness and adoption barriers	Simple farmer-centric UI
2025	Agricultural product digitalization	Literature/secondary research	Improved accessibility and efficiency	Infrastructure/digital literacy	Inclusive marketplace
2026	Buyer-seller matching	Mobile application	Direct matching possible	Requires real-world validation	Integrated scalable marketplace

7. COMPARATIVE ANALYSIS

The reviewed research demonstrates a gradual evolution of digital agricultural marketing.

Early research primarily focused on **online auctions and e-commerce adoption**. The 2007 online coffee auction study demonstrated that online procurement can connect agricultural producers with wider markets. The 2013 Trichy study further highlighted the importance of farmer adoption of e-commerce.

Later research shifted toward **integrated agricultural e-commerce platforms**. Cloud-based agricultural platforms introduced online product search, supply-demand information, purchasing, payment, and delivery-related functionality.

More recent research has focused on **digital farmgate systems, e-NAM adoption, digital literacy, and direct buyer-seller matching**. These studies show that technology alone is insufficient. Successful adoption also depends on usability, awareness, infrastructure, trust, and interoperability.

Therefore, existing approaches can be broadly classified into:

1. Online agricultural auctions
2. General agricultural e-commerce
3. Government digital market platforms
4. Mobile agricultural information systems
5. Digital farmgate platforms
6. Direct farmer-buyer marketplace applications

The proposed Digital Farmer Marketplace builds upon these approaches by focusing specifically on **direct farmer-buyer connectivity and a simple user-oriented marketplace interface**.

8 LIMITATIONS IDENTIFIED FROM EXISTING RESEARCH

The literature review identifies the following major limitations.

8.1 Digital Literacy

Many farmers may not have sufficient knowledge of smartphones, applications, or online transactions.

8.2 Infrastructure

Poor internet connectivity and limited digital infrastructure can restrict access to digital agricultural platforms.

8.3 Platform Fragmentation

Different agricultural platforms may operate independently and may not exchange information effectively.

8.4 Logistics

Agricultural products are often perishable and require timely transportation.

8.5 Trust and Verification

Online buyers need confidence regarding:

- Farmer identity
- Product quality
- Product quantity
- Price
- Transaction reliability

8.6 Limited Direct Interaction

Some systems focus primarily on market information or auctions rather than establishing a complete farmer-buyer relationship.

8.7 Adoption Barriers

Even technically capable platforms may experience low adoption because of:

- Lack of awareness
- Complexity
- Perceived risk
- Lack of training

8.8 Lack of Personalization

Existing systems may provide general market information without personalized recommendations based on location, product availability, demand, or buyer preferences.

9. RESEARCH GAP

Based on the literature survey, several research gaps have been identified.

Although digital agricultural platforms have demonstrated the ability to improve market access, existing approaches continue to face challenges related to digital literacy, platform fragmentation, logistics, trust, interoperability, and usability.

Many existing systems concentrate on individual aspects such as:

- market information,
- online auctions,
- product listing,
- e-commerce,
- government market access,
- or communication.

However, there is still scope for an integrated and farmer-friendly platform that combines these functions into a single system.

The major research gap identified is therefore:

There is a need for a simple, scalable, agriculture-specific digital marketplace that provides direct farmer-buyer connectivity while integrating product listing, transparent pricing, product discovery, communication, transaction support, and future logistics and payment services, without creating excessive complexity for farmers with limited digital literacy.

This gap forms the foundation of the proposed Digital Farmer Marketplace.

10. PROPOSED SYSTEM BASED ON THE RESEARCH GAP

The proposed Digital Farmer Marketplace aims to address the identified research gaps through a centralized web-based platform.

Farmer Side

Farmers can:

- Create an account
- Create a farmer profile
- Add agricultural products
- Specify price
- Specify quantity
- Update product availability
- View buyer interactions

Buyer Side

Buyers can:

- Register/login
- Browse agricultural products
- Search products
- Filter products
- View product information
- View farmer information
- Contact farmers
- Place orders in future versions

Marketplace Layer

The platform provides:

- Product catalogue
- Search
- Product categories

- Price information
- Availability information
- Farmer profiles

Future Integration

The system can later integrate:

- Online payments
- GPS/location services
- Delivery tracking
- Logistics providers
- AI price prediction
- Demand forecasting
- Recommendation systems
- Product traceability

11. FUTURE RESEARCH DIRECTIONS

11.1 AI-Based Price Prediction

Machine-learning models can analyze historical market prices and recommend suitable prices to farmers.

11.2 Demand Forecasting

Historical transactions, seasonal trends, weather information, and market demand can be analyzed to predict future demand.

11.3 Recommendation System

The system could recommend:

- products to buyers,
- buyers to farmers,
- products based on location,
- products based on previous purchases.

11.4 Multilingual Interface

Supporting regional languages can improve accessibility for farmers.

For example:

- Tamil
- Telugu
- Hindi
- Kannada
- Malayalam

11.5 Voice-Based Interaction

Voice interfaces could help farmers with limited literacy or difficulty typing.

11.6 Secure Digital Payments

Future versions can integrate secure digital payment systems.

11.7 Logistics Integration

The platform can connect farmers with transportation providers and support:

- pickup scheduling,
- delivery tracking,
- location sharing.

11.8 Product Traceability

Blockchain or other distributed-ledger technologies could potentially be used to maintain transparent product histories.

11.9 IoT-Based Freshness Monitoring

IoT sensors could be integrated into future supply chains to monitor:

- temperature,
- humidity,
- storage conditions,
- transportation conditions.

12. CONCLUSION

The literature survey demonstrates that digital technologies have significant potential to transform agricultural marketing by improving market access, information availability, communication, and farmer-buyer connectivity. Previous studies have explored online agricultural auctions, e-commerce adoption, cloud-based agricultural platforms, digital farmgate systems, e-NAM, mobile communication tools, and direct buyer-seller matching. These studies demonstrate the benefits of digital agricultural systems but also identify significant challenges related to digital literacy, infrastructure, logistics, trust, adoption, and platform interoperability. The review indicates that a successful digital agricultural marketplace must go beyond simple product listing. It should provide a **simple, accessible, transparent, and integrated environment** for farmers and buyers. The proposed **Digital Farmer Marketplace** addresses this identified research gap by focusing on direct farmer-buyer connectivity and providing a centralized platform for agricultural product discovery and management. Future extensions involving AI-based price prediction, demand forecasting, multilingual interfaces, digital payments, logistics integration, and product traceability can further improve the platform.

13. REFERENCES

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